

A clean alternative to PSE's gas-centric future

Prepared by Washington Clean Energy Coalition · June 2026

Summary

As required by law, Puget Sound Energy has modeled how it will serve its electric and gas customers in coming decades. Their model sets the stage for PSE's 2027 Integrated Service Plan to be published later this year.

For its electric grid, PSE proposes to build more than 5,000 MW of new gas peaker plants (approximately 20 facilities) by 2050, the largest buildout of gas plants in the company's history. For its gas system, PSE assumes it will serve about the same number of gas customers in 2050 as it does today.

This preliminary plan would produce megatons of emissions, blowing past Washington's climate goals. Our electric and gas bills would rise dramatically as PSE passes on billions of dollars of allowances to permit these emissions.

There is a better way! The Washington Clean Energy Coalition, an all-volunteer non-profit organization, carefully examined PSE's data and assumptions. Hiding within that data, we found the basis of a plan that eliminates any need for the peaker plants, risky nuclear plants, and most of PSE's gas system. To our amazement, this alternative plan would be more certain than PSE's plan without increasing our costs or destabilizing the energy grid.

The problem is that PSE did not include any clean diversified plan in its modeling. Given the threat of climate change, we ask that a clean plan be modeled and independently validated by a consultant reporting to PSE's regulator, the Washington Utilities and Transportation Commission. This is the prudent thing to do before we are charged billions of dollars to build the system PSE prefers because it maximizes returns for its shareholders.

PSE's electric plan

PSE's proposed resource acquisitions for the electric grid appear on slide 30 of its May 28 Resource Planning Advisory Group (RPAG) presentation. The table below compares two of PSE's possible plans, with Reference B being PSE's preferred plan:

Cumulative Resource Additions (MW)	Reference A		Reference B	
	2035	2050	2035	2050
Flex-fuel peakers	1,674	0	1,924	5,042
Storage	3,980	54,940	1,640	2,090
Wind	2,300	3,900	1,700	3,100
Solar	1,220	1,675	975	1,275
Nuclear	0	0	0	300
Geothermal	0	0	0	100
Total	9,174	60,515	6,239	11,907

Cumulative Supply-Side Resource Builds
B Reference vs. A Reference

Reference A relies on 54,940 MW of battery storage with no assistance from peakers, nuclear, or geothermal. PSE rightly concludes that this portfolio is unrealistic. It would be hard to buy that many batteries and site them, and it would cost over \$100 billion dollars. PSE says the portfolio “failed to solve without manual intervention.” But that failure is structural, not physical. In its June 23 RPAG recap, PSE implies that clean portfolios cannot serve growing demand reliably, because any clean portfolio would suffer the same challenges as this single-technology all-battery portfolio. This conclusion is unfounded unless PSE or an independent analyst fairly tests a more balanced clean portfolio.

That leaves Reference B, which relies on 5,000 MW of “flex-fuel peakers,” small amounts of nuclear and geothermal, and a modest amount of wind and solar. The estimated cost of \$55.4 billion is almost half the cost of Reference A, so PSE prefers it, and so do we. But the choice is forced. It’s like choosing between a dog and a cat, but the cat costs twice as much and is barely alive.

Reference B has significant drawbacks. PSE calls its thermal plants “flex-fuel peakers” because, in theory, they can burn non-emitting biogas. However, PSE’s own analysis (slide 44) finds renewable fuel supplies limited to 4–5% of the amount needed to run the peakers. If there isn’t enough biogas to supply 100% of the need, we ask PSE to explain how these peakers will comply with emissions requirements. And if they don’t comply then PSE should account for how these stranded assets will serve the customers who paid for them. For the remainder of this document, we will call them “gas peakers,” because that more accurately describes what they are.

The choice of a reference portfolio is an important foundation for the Integrated System Plan. The reference case anchors every sensitivity that follows. If peakers are the preferred solution, PSE will start buying them soon, adding 1,228 MW of new peakers by 2031.

PSE may also lean on a regional backdrop: a utility-sponsored E3 study projecting a roughly 9 GW Northwest capacity gap by 2030 and concluding that new gas peaking plays an important reliability role. That backdrop deserves the same scrutiny as the ISP itself. An independent review of the study found the gap depends strongly on highly uncertain data-center load growth, and that large-load flexibility “could mitigate most or all near-term winter resource adequacy needs under most load scenarios,” a resource the study’s first phase did not consider. The Northwest Power and Conservation Council’s most recent adequacy assessment finds the region adequate through 2029 if its efficiency, renewable, and demand-response targets are met. A regional gap built on uncertain forecasts does not relieve PSE of demonstrating its own territory-specific need.

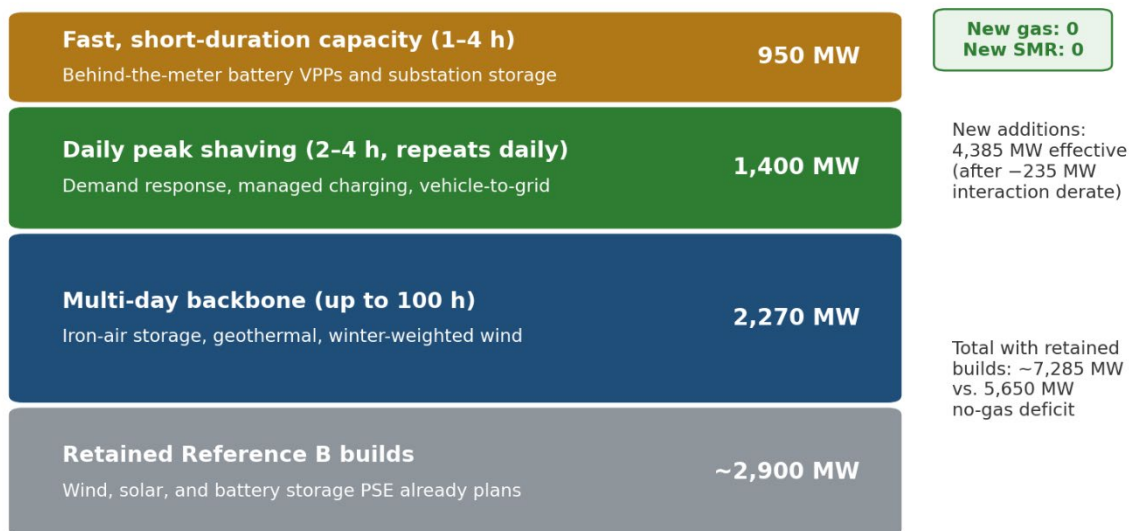
Now is the time to ask if a better portfolio is possible.

A clean electric plan, in brief

The Pacific Northwest’s hardest electric reliability event is a winter cold snap lasting two to four days, like the January 12–16, 2024 event in which the Northwest set a winter peak demand record near 35,500 MW and imported an average of roughly 4,900 MW across five days. That same event exposed the fuel system behind the peakers: on January 13, PSE’s Jackson Prairie storage facility, the region’s largest, suffered a complete outage from failed control cables, and PSE asked customers that evening to conserve both gas and electricity. Gas plants are only as firm as the gas system behind them. A credible alternative must work sequentially through such an event. We propose a portfolio in three functional layers, sized so that no resource is asked to do a job its duration cannot support. We offer a speculated architecture for testing through time intervals (not a final dispatch model).

1. **Multi-day backbone: iron-air batteries (100-hour), geothermal, and winter-weighted wind.** These resources can carry the sustained deficit across days two and three. Wind from Montana and North Dakota is included because its cold-snap output is weakly correlated with coastal stagnation events, providing recharge energy between peaks.
2. **Daily peak shaving: demand response, managed charging, and V2G.** These mechanisms compress the morning and evening demand spikes. V2G vehicles recharge overnight and can repeat daily, while DR is sized for repeated short events, never multi-day curtailment.
3. **Fast, short-duration capacity: behind-the-meter and substation batteries.** These handle the steepest one-to-four-hour needs and recharge in the valleys between peaks.

A clean portfolio in three layers, sized for a multi-day cold snap



2050 planning case, effective winter MW. Screening-level allocations from the resource table. Bands not to exact scale.

The table below gives our illustrative allocations using conservative discounts that we believe are easy to defend for accreditation today. The figures in parentheses show achievable performance consistent with documented precedents. All values are speculative planning estimates of winter capacity. These are not model-derived Effective Load Carrying Capabilities (ELCCs), and we invite PSE or a consultant to update them with formal analysis.

Note that we cautiously exclude Goldendale pumped storage and show it only as upside, due to active litigation by the Yakama Nation and Columbia Riverkeeper. If the project moves forward, it makes the clean portfolio that much easier to achieve.

The table is also cautious about demand-side and short-duration resources because they partly respond to the same hours. We conservatively apply an additional 10% discount to their combined contribution.

Resource	2035 planning (achievable)	2050 planning (achievable)	Duration	Discounts applied / location
Demand response & flexible load (includes managed V1G charging, excludes V2G)	450 (565)	600 (850)	2–4 h/event	Counts only customers who have actually signed up, with equipment the utility can verify and call on in winter. We then take 20% off the ~565 MW implied by PSE's own 10% demand-response statement (Appendix C). In-zone.
V2G (bidirectional only)	100 (250)	800 (1,500)	3 h/event, repeatable daily	Discounted at every step: how many EVs exist, how many can send power back, how many sign up, how much each provides, how many are plugged in at winter peak (Appendix C). In-zone.
Behind-the-meter battery VPPs	135 (170)	340 (425)	2–4 h	Assumes only 80% of batteries respond when called, and counts only 85% of their rated power, on 200–500 MW installed. In-zone.
Substation-sited containerized storage	460 (510)	610 (680)	4–8 h	Utility-owned batteries at substations. Counts 85% of rated power, available 90% of the time, on 600–800 MW installed. In-zone.
Iron-air 100-hour storage	335 (380)	1,900 (1,900)	100 h	Multi-day batteries, available 95% of the time in winter, on 350–2,000 MW installed, at PSE's 2035 cost entry. This is in addition to the battery storage Reference B already builds. In-zone.
Additional winter-weighted wind (MT/ND)	175 (175)	350 (350)	n/a (energy)	Counts 35% of rated power toward the winter peak, on 500–1,000 MW of new wind. The one resource that depends on transmission from out of state (Appendix A).
Enhanced geothermal (replaces 300 MW SMR)	90 (95)	400 (450)	Firm baseload	Counts 90% of rated power. PSE's table has EGS available 2035 at \$11,893/kW vs. SMR \$13,518/kW at 2040. The first 100 MW is added in 2035, the year PSE's own table says the technology arrives.
Interaction discount on demand-side & short-duration lines	–115 (–150)	–235 (–346)	—	We subtract another 10% from the first four rows combined, because resources responding to the same peak hours partly overlap.
Total (no Goldendale, no gas)	1,630 (1,995)	4,765 (5,810)		
Retained Reference B builds (wind, solar, storage)	2,040	2,900		
Whole portfolio (additions + retained)	3,670 (4,035)	7,665 (8,710)		
Upside: Goldendale pumped storage share	+540	+540	12 h	FERC-licensed January 2026 (P-14861), rehearing denied, Ninth Circuit petitions filed May 2026. Counted in no total. A conditional route to closing the 2035 gap (see text).
PSE's no-gas capacity need	3,960	5,650		What the system must add without gas. 2050 from slide 24. 2035 derived from Reference B's own builds (Appendix C). For 2045, see Appendix A.
Whole-portfolio coverage of the need	93% (102%)	136% (154%)		Whole portfolio (two rows above) as a share of the need. Goldendale-inclusive percentages appear in the text.

“Capacity credit” is the share of a resource’s nameplate (maximum) output that is dependable at the winter peak. Nameplate is discounted to effective MW by the capacity-credit and availability factors shown in this table. “In-zone” means located within PSE’s own service area, requiring no new long-distance transmission.

The table compares each portfolio in its entirety against the whole need in 2035 and 2050. Our portfolio shares every element of Reference B except the gas peakers and the SMR, which we replace with cleaner options. Counting our additions together with those shared builds, the planning case covers 93% of the need in 2035, 116% in 2045, and 136% by 2050. The achievable case covers 102%, 136%, and 154% (see calculations in Appendix C).

Although 2050 is the end date for the ISP, 2045 is the crucial year, because that is when the Clean Energy Transformation Act requires all electricity sold in Washington to come from renewable or non-emitting sources. PSE publishes no 2045 resource figures, so Appendix A tests that year directly and applies various stresses to test vulnerabilities. The portfolio clears the full need with 16% to spare and with comfortable margins for every reasonable stress test we could justify. And remember, these are deliberately conservative figures, using accreditation discounts a regulator could adopt today, applied on top of the reserve margin already built into PSE's requirement.

For our portfolio, the earliest years are the most challenging. The only year the planning case falls below 100% is 2035, and the gap is about 300 MW. Three independent paths address the shortfall: the 800 MW contingency reserve held in Appendix A, achievable-case program performance (102%), or the 540 MW Goldendale pumped storage project if its license survives review (106%, adds roughly \$1.1–1.3 billion to our portfolio cost based on the project's published \$2.5–2.9 billion construction estimate). And if elements slip beyond what the contingency reserve covers, the final backstop is the one PSE holds today: a single, deferred peaker procurement, decided around 2030 with several years of real program performance in hand, sized to the actual residual instead of a forecast (Appendix A). We do not prefer that outcome, but a portfolio that needs only one peaker to serve an extreme scenario is more prudent than a plan that commits to twenty peakers up front.

Is a clean electric portfolio economically feasible? Independent analysis supports our position that a clean portfolio is feasible. Brattle finds a virtual power plant delivers resource adequacy at roughly 40–60% of the net cost of a gas peaker. Substituting PSE's own unit costs to replace peakers and SMRs with a clean resource mix lands in the same broad range as Reference B:

Electric plan NPV, 2028–2050	Reference A (rejected)	Reference B (PSE)	WCEC clean (speculated)
Total electric portfolio cost	\$100.1B	\$55.4B	~\$55B (±\$5B)
Shared base, retained in both plans	—	~\$39B (retained storage, wind, and solar, plus conservation, DR, market purchases, and system planning)	~\$39B (identical, the same retained resources)
Acquisition costs (capital)	—	~\$12.3B (5,042 MW peakers + 300 MW SMR)	~\$13.9B (iron-air, storage, wind, geothermal, chargers)
Operating costs	—	~\$3.7B (fuel + O&M + CCA allowances at ceiling)	~\$2.0B (customer payments + charging energy)

PSE's gas plan

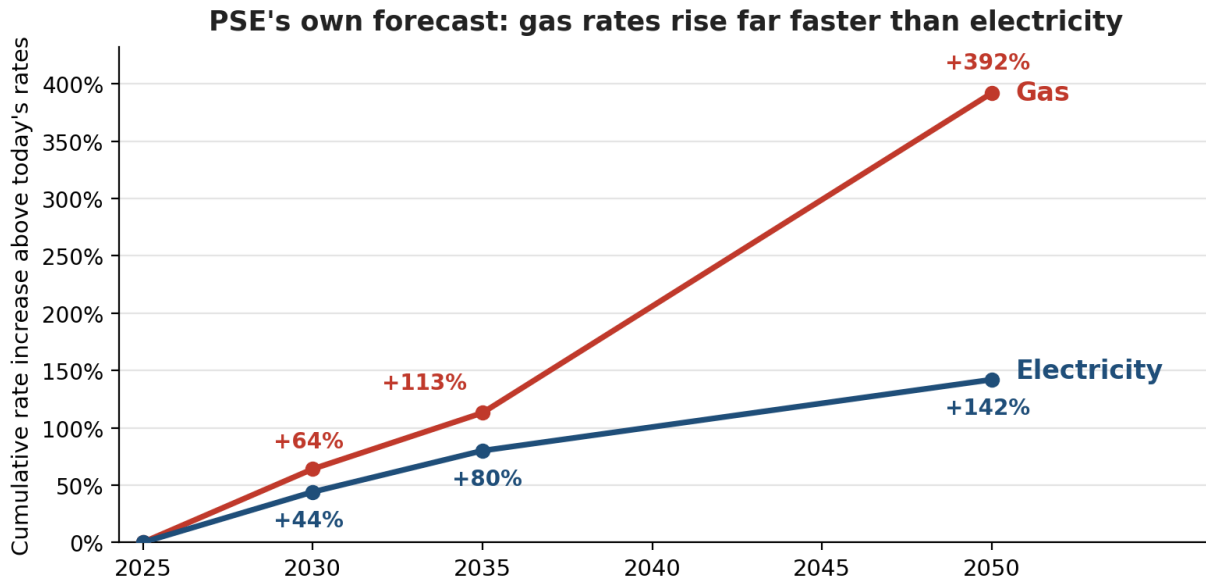
PSE's gas plan holds the gas network physically constant: "miles of main remains constant," pressure-regulating stations constant, and so on. Electrification seems like an afterthought, "limited to 3–5 small projects annually." Demand-side electrification is "limited to the existing Duvall pilot program." PSE would comply with the Climate Commitment Act not by reducing emissions but by buying allowances and renewable-fuel certificates to cover its continuing emissions. This approach is difficult to reconcile with the Act's purpose. The ever-increasing costs are simply passed on to customers.

As enacted in 2024, HB 1589 required PSE's integrated plan to "achieve all cost-effective electrification of end uses currently served by natural gas," including geographically targeted electrification. Initiative 2066 struck that language in December 2024 and barred the Commission from approving a plan that "requires or incentivizes" terminating natural gas service. A King County court has since ruled I-2066 unconstitutional. The question is now before the Washington Supreme Court (argued January 22, 2026), and the original requirements revive if that ruling is affirmed.

No matter how the case is resolved, what survives the passage of I-2066 is sufficient. The combined plan must still address emissions, customer costs, reliability, and equity. It must still include low-income electrification programs (RCW 80.86.020(4)(h)). And the Commission must still find the plan to be the lowest reasonable cost path. Nothing we request requires or incentivizes terminating gas service. We ask only that PSE model the customer-driven attrition that is already economically plausible, and plan equitably for it. Whatever the Court ultimately decides, I-2066 does not prohibit modeling, and prudent planning requires it.

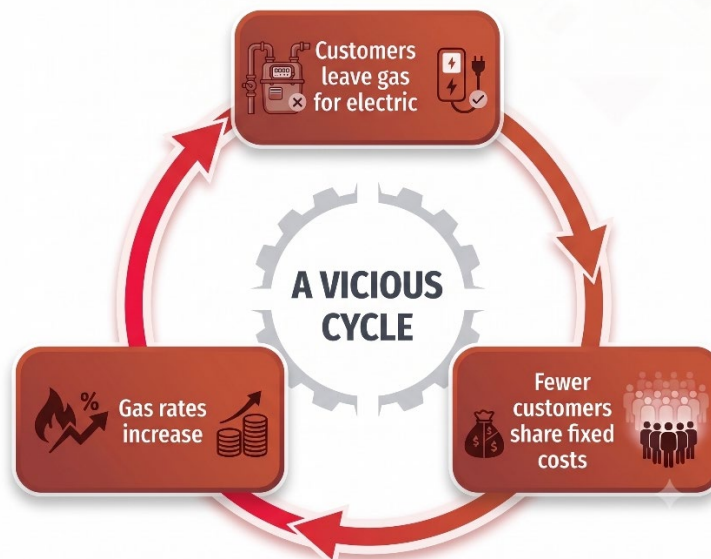
Our main concern is that PSE's **flat forecast is probably wrong and risky for customers**. Several factors are likely to push gas-customer attrition well beyond PSE's reference assumptions:

1. PSE expects the price of gas to rise more quickly than the price of electricity (see graph below).
2. Fossil fuels can experience unanticipated price shocks due to issues such as global demand and geopolitical events involving major oil and gas producers.
3. Heat pumps are roughly three times as efficient as gas furnaces. Many customers are likely to lower their monthly energy bills by converting, especially where incentives, weatherization, and cold-climate equipment are available.



Source: PSE 2027 ISP reference case (RPAG, May 28, 2026), slides 49 and 51. Illustrative; assumes CCA ceiling prices.

All three of the factors listed above will incentivize gas customers to switch to electricity to save money. But PSE's gas planning assumptions (slide 59) ignore these factors, projecting that customer load "remains stable in the near term, followed by an approximate 10% decline by 2050," while holding customer count "constant over planning horizon." If this forecast is wrong, and if higher-income households leave gas faster than renters and low-income households, then the latter will have to shoulder the costs of maintaining gas mains, meters, and other gas infrastructure. As those fixed costs are distributed across a declining customer base, gas bills will rise, and more customers will be pushed to leave. This process feeds on itself, a vicious cycle that falls hardest on the customers least able to escape it.



Even after I-2066, RCW 80.86.020(4)(h) still requires PSE to operate low-income electrification programs (heat-pump incentives, energy-burden mitigation, and bill assistance). We believe anticipating these problems and proactively addressing them is in everyone's best interest.

Risks for PSE

The vicious cycle described above is a major problem for customers who are left behind. But it is equally problematic for PSE. Regulators often allow the utility to keep recovering its fixed costs from whomever remains (Davis and Hausman, 2022). Each round of recovery raises bills, and higher bills push out the next tranche of customers. PSE doesn't experience too much pain as long as the Commission keeps approving rate increases on the shrinking base of customers. But if the Commission eventually declines to allow PSE to collect ever-higher rates on the smaller, lower-income base that remains, the undepreciated gas system becomes a shareholder loss.

For customers worried about their bills, electricity is not the only exit. Propane is delivered throughout Washington today, and about 4% of Northwest homes already heat with it. Propane is essentially unregulated, and most local dealers fall below the Climate Commitment Act's 25,000-ton coverage threshold. Propane therefore carries no allowance cost, while every therm that PSE delivers does. The Legislature has already seen this gap being exploited: a 2026 law lowered the threshold to 500 tons after fuel distributors formed new companies deliberately designed to stay under the cap. However, the law applies only to suppliers created after January 2023 and leaves incumbent dealers unaffected. Today, PSE's gas is still about half the cost of propane per unit of heat. But PSE's gas rates rose 10.6% in January 2025 and about 7% more in January 2026, and residential gas deliveries fell about 7% from 2022 to 2023. Every increase narrows the price gap toward a fuel that pays no carbon price at all.

Many gas utilities treat this as a real risk when they communicate with investors. The SEC filings for NW Natural, the region's largest pure-play gas utility, state that its distribution business competes "primarily with suppliers of electricity, fuel oil, and propane" and that high gas prices "may negatively affect our ability to secure new customers or retain our existing customers." Credit-rating agencies have flagged gas distribution as a sector with growing transition risk. And when SoCalGas permanently shut off service to a landslide-prone California neighborhood in 2024, homes were running on propane within weeks.

Propane defection would be the worst outcome for every party. The customer burns a fuel that emits about 19% more CO₂ per unit of heat than natural gas (EPA emission factors: 62.9 versus 53.1 kg per MMBtu). The state loses the carbon price on that heat entirely. And PSE loses the customer just as surely as if they had electrified, with nothing gained for the climate. A plan that holds customer count "constant over planning horizon" leaves this entire dynamic unexamined and exposed to chaotic defection. We ask PSE to model rate-driven attrition with both exits open, electrification and unregulated fuels. It is important to keep heating load inside a regulated, decarbonizing system rather than leaking it to a dirtier, unregulated one (Request 9).

Scaling the electric plan for full electrification

Can the electric grid handle increased electric loads if every gas customer converts to electricity? (We model residential conversions only. Commercial gas load is a separate increment PSE should add.) If approximately 800,000 homes need electricity for heating during the peak hour of a cold snap, that would require an additional 2,400–3,600 MW. That adds roughly 50% to the ~5,650 MW peak requirement PSE already models. A load of this size appears largely absent from PSE’s electric forecast. The company does not model this load, because its reference assumptions hold gas-customer attrition to a very low level for the next couple of decades.

Other possibilities deserve study. If only 400,000 conversions from gas to electricity occur, the added peak is roughly half as large. On the other hand, higher loads could materialize due to faster conversion, weaker weatherization, or heat pumps defaulting to electric-resistance backup in a deep freeze. We use the mid case for illustration and ask PSE to model the range.

Scenario	Converted homes	Gross peak (MW)	– Efficiency	– Flexibility	Net firm need (MW)
Low	~400,000	~1,400	–350	–475	~575
Mid	~800,000	~2,800	–700	–950	~1,150
High*	~800,000	~3,600	–540	–1,060	~2,000

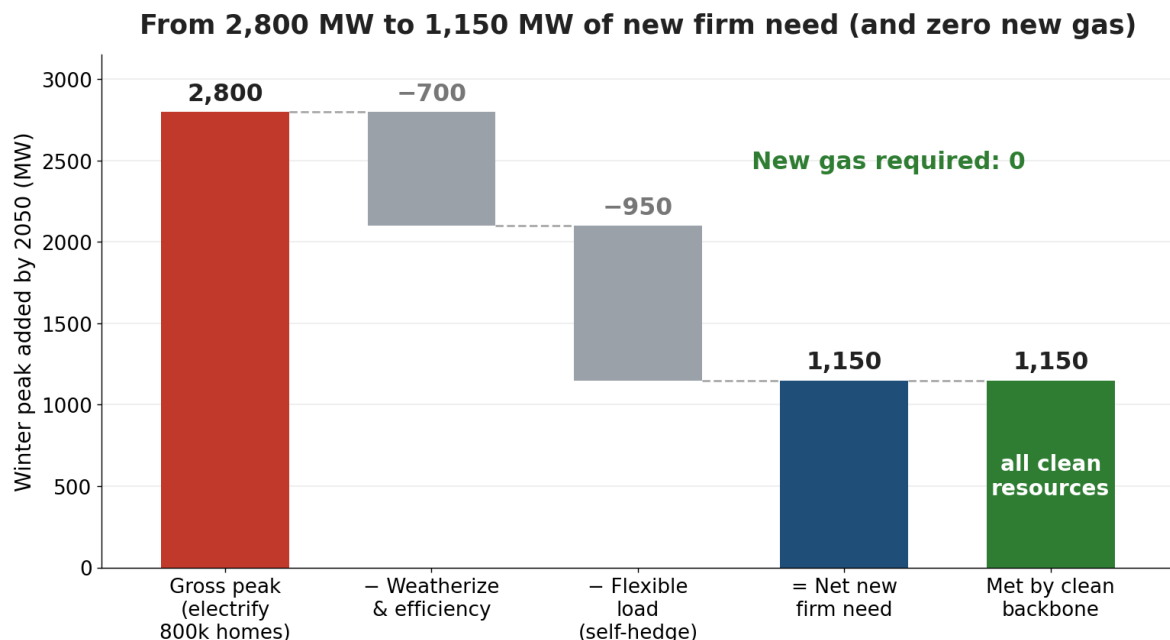
*High case: we assume things go badly. Homes draw at the top of the range (4.5 kW instead of 3.5), weatherization saves less (15% instead of 25%), and we trust the flexibility programs less, crediting about 35% of the remaining peak instead of 45%, because in a deep freeze more customers are more likely to override control events and more heat pumps might fall back to power-hungry electric-resistance heat. But even in this case, the answer is to build more of the same clean resources (more long-duration storage and geothermal, plus Montana wind as transmission allows). We leave the emergency reserve in Appendix A untouched, since it is set aside for construction delays, not for bigger loads. The best defense is to keep the high case from happening by supporting building codes and incentives that steer people toward cold-climate heat pumps, which rarely need resistance backup.

For all the scenarios, the added load is better matched to efficiency, flexibility, long-duration storage, and firm clean resources than to a large fixed buildout of gas peakers. The electrified appliances are among the most controllable loads on the system. For example, a heat-pump water heater is a thermal battery, and a well-insulated home is thermal storage that can be warmed a degree or two ahead of the peak, then coasts through the peak on stored heat. We call this “heat-pump pre-heat.” Our portfolio performs better because a substantial share of the added load is controllable and efficiency-sensitive rather than fixed. Of the ~950 MW of flexibility included in the mid case, we counted only load that could be contractually enrolled or available through controls-enabled appliance standards. These consider comfort and health protections that limit how long any device is expected to participate.

The single largest risk in this scenario is a heat pump that switches to electric-resistance heat at very low temperatures. Heat pumps designed for colder climates retain most of their capacity in a deep freeze, but a home that falls back to resistance heating can draw 10–15 kW instead of 3–4 kW. To avoid wasteful and expensive problems of this type, we need targeted incentives, building codes, and weatherization to steer conversions toward cold-climate equipment. Even so, PSE should model the resistance-backup case directly.

Other observations: Weatherization paired with conversion trims the heat load by roughly a quarter. A multi-day freeze is exactly why 100-hour storage and firm resources are included in our portfolio. Four-hour batteries can't cover this scenario alone, which is the flaw identified in PSE's rejected Reference A. And while gas peakers can run for days, they do so only by burning fuel continuously through the event, producing emissions that are restricted under the Clean Energy Transformation Act.

Full-electrification winter-peak increment by 2050 (speculated)	MW
Gross unmanaged peak added by converting ~800,000 residential gas homes to heat pumps (cold-snap hour)	~2,800
Less: weatherization and efficiency as homes convert (cost-effective measures only)	-700
Less: self-managed flexibility from the electrified loads (smart water heaters, heat-pump pre-heat, managed charging)	-950
= Net new firm capacity required by 2050	~1,150
Met by expanding the clean backbone: +500 iron-air, +300 winter wind, +200 geothermal, +150 substation/BTM	1,150
New gas required	0



Screening mid case (full residential electrification). See memo Appendix for the low/high range.

The table and graph above show how a gross ~2,800 MW peak due to widespread electrification can be reduced to about 1,150 MW of new firm capacity. Again, this is a speculated mid-case estimate, not a forecast. Weaker weatherization, lower program participation, or reliance on heat pump resistance backup would raise it materially. But it can also be met by scaling the same clean backbone. The table below shows how our portfolio preserves the margin before and after electrification.

2050 winter capacity ledger (planning case, speculated)	Before electrification	With full electrification
Base need: PSE's 2050 no-gas deficit (slide 24, flat-gas forecast)	5,650	5,650
Added firm need PSE does not model (~800,000 conversions)	—	+1,150
Total need (MW)	5,650	6,800
New clean additions (this portfolio)	4,765	4,765
Backbone expansion for electrification	—	+1,150
Retained Reference B wind, solar, and storage (effective)	~2,900	~2,900
Total supply (MW)	~7,665	~8,815
Margin (MW)	~+2,015	~+2,015

Electrification adds 1,150 MW to both the need and the supply, so the margin survives intact. New gas does not appear to be necessary in this speculated case. Building gas plants to serve electrified heat would be self-defeating in any case: it merely moves the gas from basements to power plants, keeps paying carbon allowances, and strands the peaker plant asset at the 2045 100%-clean deadline.

Our report does not itemize additional distribution reinforcement such as the transformers, feeders, and service upgrades that widespread electrification will probably require. Recent national modeling estimates distribution reinforcement for full residential electrification at roughly \$2,800–6,400 per household, and finds that demand-side management eliminates over two-thirds of that cost, bringing it to roughly \$900–2,100. Applied to ~800,000 conversions, that suggests roughly \$2–5 billion in PSE's territory unmanaged, falling to roughly \$1–2 billion with the managed charging, heat-pump pre-heat, and distributed storage already central to this portfolio. Three further considerations temper the gross figure. 1) The cost is partly offset by the gas mains, services, and pipeline-integrity investment PSE avoids as customers leave gas. 2) The managed charging, heat-pump pre-heat, and distributed storage in our portfolio flatten the local coincident peaks that drive distribution upgrades. 3) Substation-sited storage can serve as a non-wires alternative that defers specific feeder and transformer work.

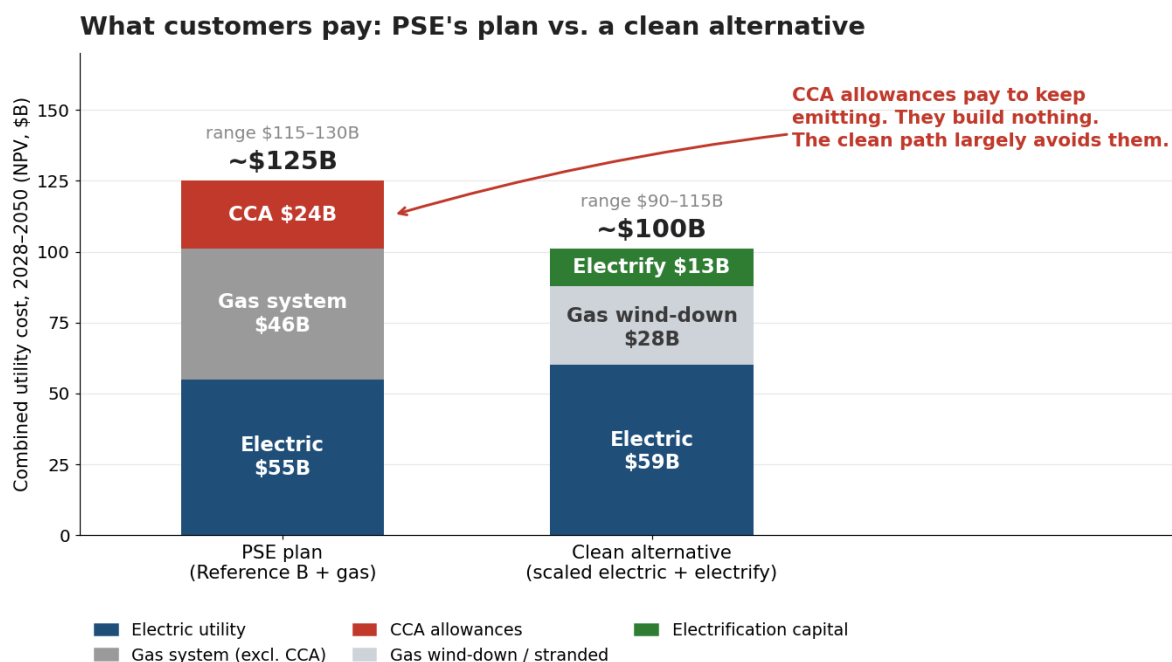
In the end, distribution reinforcement is a cost of electrification that any HB 1589-compliant plan must bear, and PSE should be required to run high-electrification feeder-impact studies that incorporate these non-wires alternatives rather than defaulting to traditional upgrades. On the evidence available, the net cost appears material but not obviously prohibitive.

What a clean portfolio costs and saves

As we have shown, a clean portfolio serving the electric side alone can achieve rough cost parity with PSE's preferred plan. What happens to costs if high rates of electrification cause electric demand to increase significantly?

PSE's flat-gas plan would spend an enormous and growing sum on Climate Commitment Act allowances simply to keep the gas system emitting. By 2050 those allowances would be the majority of each gas bill: money that buys no pipe, no appliance, and no clean energy. Avoiding allowance payments doesn't give us a magic cash account that can simply be repurposed. But within an integrated planning framework, avoided allowances create fiscal flexibility through mechanisms like transition financing, securitization, and rate design. These support electrification investment that would otherwise look unaffordable. For an individual household under these assumptions, a transition plan might finance conversion at a level that costs a customer less than the cumulative gas surcharge on the flat-gas path.

Illustrative combined-utility scenarios (2028–2050 NPV, not a revenue-requirement result)	PSE path	WCEC path
Electric plan (utility cost)	\$55.4B (Reference B)	~\$59.5B (clean, scaled for electrification)
Gas plan (utility cost)	~\$70B (gas supply \$29.2B plus pipes and delivery plus carbon allowances, with allowances reaching 51% of gas cost by 2050)	~\$20–40B (remaining gas service plus stranded pipes during a managed wind-down, scenario-dependent, carbon allowances largely avoided)
Electrification capital (customer-side)	— (gas system stays intact)	~\$10–16B (\$12–20k/home incremental × 800k)
Combined total	~\$125B (~\$115–130B)	~\$90–115B



Screening estimate with wide, overlapping error bands. Not a revenue-requirement result. See memo Appendix.

The table above deliberately blends utility system cost (the electric and gas rows) and customer-side transition capital (the electrification row) to provide an integrated view, not a commission-grade revenue requirement. The residual gas figure is scenario-based and the electrification capital is a range. With those caveats, the electrification capital that might look like a dealbreaker is comparable to the gas costs it avoids. When CCA allowances are also considered, the costs of the clean portfolio are plausible enough to merit careful modeling. Even if the two plans had similar costs, the clean plan has two further advantages not shown in the table. A converted home's delivered energy falls by roughly 60%, so monthly bills are likely to fall for most households (especially with incentives, weatherization, and cold-climate equipment). Rebate programs reduce the conversion capital further (Washington's CCA-funded HEAR program, operating now, and the federal Home Energy Rebates of up to roughly \$14,000 for income-eligible households, pending launch in Washington).

Who pays for electrification? Today PSE's gas and electric utilities are separate rate bases with separate customer classes. As gas customers become electric customers (perhaps served by a different utility), the undepreciated value of gas distribution assets (the "stranded cost") has to be recovered from someone. Continuing customers can't be charged without risking a shrinking-system cost shift (the "vicious cycle" we described earlier). If departing customers must pay an exit charge, that could also spark an exit spiral as customers race to leave before the charge is mandated. It is difficult to ask electric-only customers to pay for gas system maintenance; charging customers for a service they don't use isn't how utility rates work.

We do not claim an easy answer to this conundrum. The combined plan is where an equitable transition and allocation of costs must be worked out, using tools such as depreciation schedules, targeted retirement, securitization, and rate design, and with explicit protection for low-income families and renters (Request 6).

Timing favors patience. Regulators will soon confront the decision to commit roughly \$3.1–3.8 billion to the first 1,924 MW of PSE's proposed peakers (the near-term mix already includes 1,228 MW by 2031). In contrast, the clean resources deploy incrementally and can be evaluated tranche by tranche. Once a peaker is built, capital and CCA exposure persist for thirty years no matter what clean technology costs over that period. If the clean portfolio underperforms, PSE has better information to procure a gas plant that is sized and scheduled to match actual need. But if PSE builds 5 GW of gas and the alternatives perform well, ratepayers own a stranded fleet. There is a difference: being partly wrong about modular clean resources is more recoverable than overcommitting to a five-gigawatt gas fleet.

Who pays for stranded gas plants? CETA requires 100% clean (renewable or non-emitting) electricity by 2045. When that year arrives, a peaker plant built in the early 2030s is only one-third to one-half depreciated (on a 30-year book life). Facilities built later would have even less depreciation. Unless a peaker can burn 100% renewable fuel (PSE's analysis caps the renewable fuel at 4-5% of the volume needed – see slide 44), the plant can't lawfully serve Washington load on fossil gas after 2045. We ask PSE to state plainly who absorbs the undepreciated balance – shareholders or ratepayers – and to include the amount in its cost comparison with a clean portfolio (Request 4).

Siting is an unpriced risk. Reference B's 5,042 MW implies roughly twenty peaker units, each needing a site, local land-use approval, and an air permit. No new gas-fired power plant has come online in Washington since 2008, when Mint Farm and Grays Harbor began operation. No application to site a peaker plant has been filed with the state siting council since the mid-2000s. The last three certified gas projects were cancelled by their developers, and the Centralia coal-to-gas conversion PSE contracted for in December 2025 repowers an existing site rather than permitting a new one.

The siting council's mandatory jurisdiction begins at 350 MW, leaving individual peakers to local permitting, community by community. CETA adds an equity test: utilities must ensure "the equitable distribution of energy and nonenergy benefits and reduction of burdens to vulnerable populations and highly impacted communities" (RCW 19.405.040(8)). Moreover, peer-reviewed analysis of California peakers found that over a third of some units' generation occurs on days when air quality already violates federal standards. None of this schedule, cost, or equity risk seems to be included in PSE's conception of Reference B. Our portfolio relies on batteries, demand response, and customer-sited equipment that have much less stringent siting requirements.

Fuel firmness is a second unpriced risk. A gas peaker is only as reliable as its fuel supply on the coldest days, when gas demand peaks and delivery can fail—exactly what happened on

January 13, 2024, when the region's largest gas-storage facility, Jackson Prairie, went fully offline and PSE asked customers to conserve. Keeping peakers firm through such events generally requires on-site backup fuel storage, dual-fuel capability, or firm pipeline contracts, each carrying cost. PSE's \$1,624/kW "Natural Gas Peaker" figure appears to cover the generating unit itself; if it omits these fuel-assurance measures, a firm peaker fleet costs more than modeled. We do not estimate that cost here, but ask PSE to disclose whether and how it is included (Request 4).

Why no nuclear?

Our portfolio substitutes enhanced geothermal for the 300 MW small modular reactor (SMR) in Reference B. This is not a judgment about nuclear power in general. It is an application of the standard Washington law sets for this plan. The integrated system plan must meet system needs at the **lowest reasonable cost and risk to the utility and its customers** (RCW 80.86.020). The statute behind utility resource planning defines lowest reasonable cost to include resource uncertainties and “the risks imposed on the utility and its ratepayers” (RCW 19.280.020). How does an SMR rate on these metrics of cost and risk?

Cost. Both nuclear and geothermal energy check the same box: non-emitting, firm capacity. PSE’s resource adequacy materials credit nuclear at a 99% Effective Load Carrying Capability (ELCC – see 2023 Electric Progress Report, Table 7.19). PSE has published no ELCC for geothermal, but firm geothermal is conventionally accredited like other baseload resources. We use a conservative ELCC of 90%. On PSE’s slide 81, geothermal arrives first and costs less than nuclear per kilowatt: enhanced geothermal is projected to cost \$11,893/kW in 2035, while the SMR arrives in 2040 at \$13,518/kW, five years later and 14% more expensive. Reference B recognizes the feasibility of geothermal by selecting 100 MW by 2050. PSE’s draft leveled screens (slides 88 and 89) embed performance, lifetime, and operating assumptions that have not been disclosed. We ask PSE to publish those assumptions along with both ELCCs from the 2027 Resource Adequacy study (Request 8). The comparison deserves scrutiny because the real-world evidence does not support choosing an SMR over geothermal.

No commercial SMR operates today in the United States. The one project that reached contract stage, NuScale’s Utah project, saw its subsidized target price rise from \$58 to \$89 per MWh before it was canceled in 2023 for lack of subscribers. The first SMR actually under construction in North America, at Ontario’s Darlington site, is budgeted at about C\$20.9 billion for 1,200 MW, near PSE’s assumed capital cost before any first-of-a-kind overrun. Lazard’s 2025 leveled costs put unsubsidized nuclear at \$141 to \$251 per MWh compared to \$66 to \$109 for geothermal. Geothermal’s trajectory is promising: Fervo’s Cape Station holds 500 MW of signed utility PPAs with project financing closed, drilling costs there have fallen by half, and NREL finds roughly 106 GW of potential for Enhanced Geothermal Systems in the Cascades. Since the 2025 federal budget law, the two technologies carry identical tax-credit phasedowns.

Risk. The SMR has risks that geothermal does not have. SMRs run on a specialized fuel known as High-Assay Low-Enriched Uranium (HALEU). U.S. facilities currently create only 900 kilograms of HALEU per year, compared to a projected annual need of 40 metric tons by 2030. Russian imports could help fill the gap, but such imports are banned starting in January 2028.

Spent fuel from an SMR has nowhere to go, and peer-reviewed work finds SMR designs can produce 2 to 30 times more waste per unit of energy. Industry disputes the finding (PNAS, 2022), so we aren’t sure who to believe. The licensing regime for a 2040 reactor does not exist yet. Presidential executive orders in May 2025 directed the Nuclear Regulatory Commission to reorganize and rewrite its rules within 18 months, amid a commissioner’s removal and heavy staff losses. Whatever one thinks of that policy, it adds schedule and safety-assurance uncertainty to any SMR commitment made today.

PSE’s draft sensitivity list already includes both SMR and geothermal as resources in the reference case. We ask that both be tested honestly: publish the assumptions, model a geothermal-first portfolio that replaces the SMR, and let the lowest reasonable cost and risk standard decide. A resource available five years sooner, at lower capital cost, with commercial momentum and none of the fuel, waste, or regulatory risks, appears hard to beat.

Actions requested from PSE

Throughout this memo we have asked PSE for specific tests, disclosures, and modeling. Here we combine all of our requests. None of our requests requires PSE to adopt our portfolio. Each asks only that the clean and electrified alternatives be evaluated as rigorously as the gas-and-nuclear plan PSE prefers.

1. **Model a diversified clean portfolio as a true third reference case.** Replace the disqualified single-resource Reference A with the diversified clean portfolio described here, or any reasonable variation of a portfolio without new gas or commercially-unproven SMRs. Run it through the same chronological reliability and production-cost models as Reference B, considering impacts on both the gas and electric systems before any preferred portfolio is selected.
2. **Model a high-electrification future as a matter of prudent planning.** Test the electric capacity need under significant levels of electrification, not only the flat-gas forecast. Include commercial gas load, a reasonable low/mid/high adoption range like ours, and heat pumps that revert to electric resistance heating in cold weather. Customer-driven attrition is plausible under any legal regime, and modeling it violates nothing in I-2066.
3. **Publish symmetric, comparable assumptions.** Give every clean resource the same paper trail the gas peakers get. For each non-gas resource, publish the values that determine how much winter capacity it is credited with (accreditation, ELCC, deliverability, duration, and enrollment). Publish the biofuel-conversion assumption behind the 5,042 MW of peakers, which is hard to square with PSE's own finding that renewable-fuel supply is limited to 4–5% of conventional fuel (slide 44). Also publish the demand-side growth embedded in the reference cases, along with PSE's ELCC scaling curves that show how a resource's capacity credit falls as more of it is added.
4. **Compare the combined economics transparently.** Produce a combined-utility revenue-requirement comparison that contrasts the CCA allowances the gas plan would pay with the customer bill savings electrification delivers. Allowances should be priced at the ceiling for all peaker dispatch, matching how PSE's reference case already prices CCA across both its gas and electric systems (slide 87). Include a stranded-cost scenario assuming the 2045 biofuel conversion fails, and disclose the assumed siting and permitting timelines for the roughly twenty peaker units Reference B requires.

In its June 23 RPAG update (slides 24-25), PSE says ~\$1.5B will be needed to integrate 3,000 MW of clean resources. PSE should compare that with the local-network-upgrade cost to interconnect peaker units and provide fuel-assurance investments to keep them firm when the gas system itself is stressed (on-site backup fuel storage, dual-fuel capability, or firm pipeline contracts). As the January 2024 Jackson Prairie outage showed, a gas peaker is only as firm as the delivery system behind it.

5. **Account for the net distribution cost of electrification.** Model the transformer, feeder, and service upgrades a high-electrification scenario requires. Account for the gas distribution investment avoided as customers leave gas and the non-wires value of distributed storage and managed load. PSE's June 23 distribution reference case already tests substation battery co-location as a non-wires alternative, but it pre-screens just 10 candidate substations and assumes only 3 of them succeed (a 30% rate). PSE should disclose the criteria that excluded the other substations and justify the 30% success rate, which looks conservative as distributed-storage technology matures (June 23 RPAG, slides 14–18).

6. **Plan an equitable gas transition.** Prepare a geographically targeted electrification and cost-allocation plan that protects renters and low-income customers from a death-spiral cost shift and specifies how stranded gas-distribution costs are recovered.
7. **Treat transmission limits as variables, not constants.** Assess dynamic line ratings, advanced reconductoring, and power-flow control on the WA East and Montana paths, as FERC Order 1920 contemplates, and disclose how much of PSE's 746 MW of Colstrip transmission rights and its prospective ~750 MW (up to 25%) North Plains Connector share is committed to Reference B's own builds versus available for winter-weighted Montana and North Dakota wind.
8. **Justify the SMR against geothermal under the lowest reasonable cost and risk standard.** Publish the ELCC, capacity-factor, lifetime, fuel, and O&M assumptions behind the nuclear and geothermal screens on slides 81, 88, and 89, and model a geothermal-first portfolio that replaces the 300 MW SMR before any nuclear commitment is made.
9. **Model customer attrition from gas with both exits open.** Replace the constant-customer-count assumption with rate-driven attrition scenarios that include defection to unregulated fuels such as propane. Report the resulting rate, revenue, and emissions-leakage effects on the gas system.

Conclusion

We are a volunteer coalition, and it should not be up to us to create a clean energy plan. A regulated monopoly with hundreds of staff and sophisticated models is far better resourced to test whether a diversified clean portfolio can serve winter peak in the Puget Sound. The integrated system plan statute and the Commission's 2025 planning rules require PSE to demonstrate that its plan is the lowest reasonable cost path, a demonstration that is impossible without testing a serious clean alternative. Our memo exists only because PSE's modeling did not seriously test such a portfolio, and instead pitted gas against a single, narrow alternative that PSE itself describes as failing to solve without manual intervention.

PSE is subject to "cost-of-service regulation," a common method used by public utility commissions to determine fair prices for monopoly services. Under cost-of-service regulation, PSE earns an attractive rate of return on infrastructure it builds and places in its rate base. Roughly twelve billion dollars of new gas turbines and nuclear capacity would deliver higher shareholder returns than demand response, vehicle-to-grid, and customer-owned batteries. Cost-of-service regulation simply tilts the field toward large, utility-owned bulk infrastructure, inherently disadvantaging the distributed, customer-owned, and demand-side resources that reduce the overall rate base.

When a company preparing an analysis has a financial stake in the outcome, sound governance requires independent verification rather than reliance on that party alone.

Our most important recommendation is directed not at PSE but at the Commission. Before approving a major, long-lived fossil buildout, the Commission should require independent review and validation of PSE's key assumptions and sensitivities, for example by directing PSE to fund an independent technical consultant selected by and reporting to Commission Staff. The consultant should run a diversified clean-and-electrified reference case within the docket, as is common in contentious proceedings. Modeling and evaluation should occur with the same rigor as PSE's preferred plan, with every assumption published. This does not displace the planning role of PSE or the Commission. It adds the independent verification and public transparency that ordinary prudence requires when the analyst has a financial interest in the result.

Deficiencies in PSE's current stakeholder process heighten the urgency of our recommendation. The Resource Planning Advisory Group (RPAG) is meant to bring outside perspectives into PSE's analysis, but PSE allows no public questions or discussion during RPAG meetings. Robust technical discussions were common in advisory group meetings for at least a decade before PSE insisted that participants be vetted and hand-picked by PSE. PSE allows written comments from the public, but deadlines are too short for detailed written analysis like this memo.

The stakes are too high to let key assumptions go untested. As matters stand, a single party sets the planning criteria, designs the models scored against them, and runs the evaluation on data and tools that outside reviewers cannot fully see. That alone is reason for the Commission to require independent verification and full publication of assumptions before the 2027 Integrated System Plan is approved. We ask only that the clean alternative be given the same fair test PSE gave its own preferred plan, and that the test be one others can check.

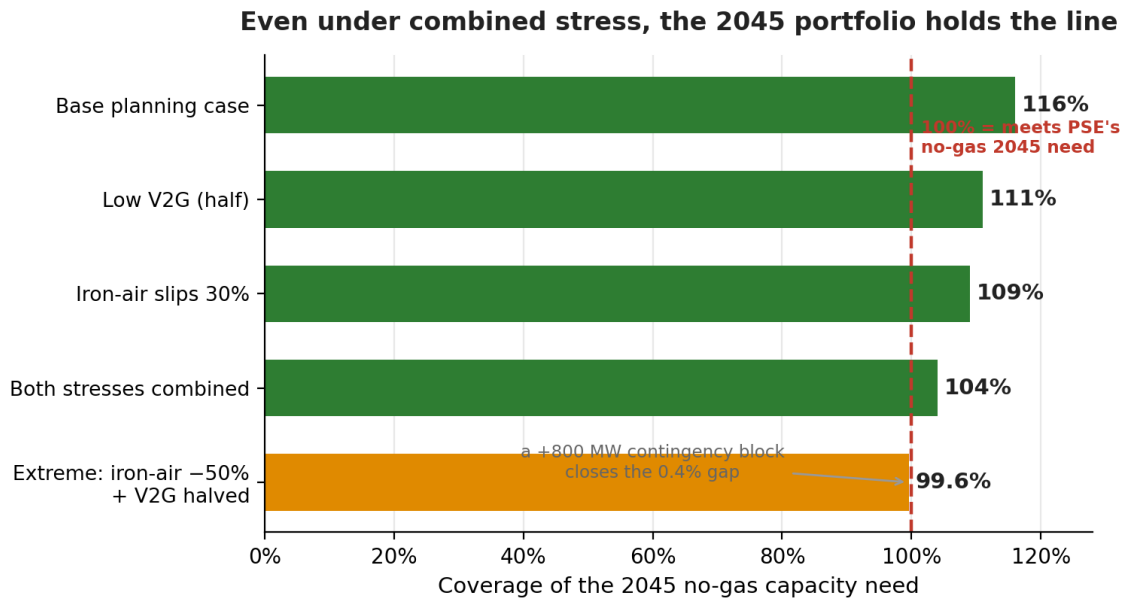
Appendix A: Testing the electric portfolio

The allocations for our proposed electric portfolio (the replacement for PSE's Reference A portfolio) are anchored by three points.

First, all of Reference B's non-gas builds (its 1,640–2,090 MW of utility-scale storage, plus wind and solar) are retained unchanged. The clean portfolio replaces only the peaker and SMR lines. Every table entry is incremental to PSE's existing ~650 MW demand-side baseline and to whatever demand-side growth Reference B embeds (Request 3 asks PSE to publish that trajectory). Six of eight resource lines are in-zone or distribution-connected, needing no new bulk transmission. The 2045 comparison point (5,126 MW) is PSE's stated deficit without gas, met by this portfolio's additions plus the retained Reference B builds (~2,705 effective MW, derivation in Appendix).

Second, the plan meets the need in 2045, the most difficult year in the planning period due to the Clean Energy Transformation Act's 100%-clean deadline. The 2045 table below uses the conservative planning numbers (a 3,220 MW interim portfolio, built from the resource trajectories in Appendix C) and then tests them under different stress conditions:

2045 case (planning numbers + retained Reference B non-gas builds)	Effective MW	vs. 5,126 MW no-gas deficit
Base planning case (3,220 + ~2,705 retained, see calculations in Appendix)	5,925	116%
Low V2G (half of planning value)	5,675	111%
Iron-air deployment slips 30%	5,583	109%
Both stresses combined	5,333	104%
Extreme case: iron-air slips 50% and V2G halved	5,105	99.6%
Contingency block held in reserve (excludes iron-air, so it hedges rather than compounds that risk): +150 MW DR, +150 MW substation storage, +200 MW contracted seasonal capacity, +300 MW accelerated wind or geothermal. Final backstop: a deferred, scaled-down thermal procurement in the late 2030s.	+800	any single element closes the extreme-case gap



Even with V2G halved and iron-air slipping 30% together, the conservative case covers 104% of PSE's stated no-gas deficit. In the extreme case the gap is 0.4%, closed by any single contingency element. A portfolio that holds margin under simple stress at the hardest milestone is worthy of formal modeling. The capacity that exceeds the need provides cushion in case there are any difficulties in procuring supply-side resources such as iron-air batteries or V2G, for example, or other resources. It covers challenges on the demand side as well, if cold-climate building codes or equipment standards are delayed or weaker than we expect.

Third, transmission is a constraint, not a blocker. Most of the portfolio does not depend on increased bulk transmission. The in-zone and distribution-connected resources (noted in the resource additions table on page 4) reduce the need for more electricity to flow across constrained transmission lines as demand peaks. The winter wind resource (Montana and North Dakota) is the only exception. Transmission of this winter wind seems feasible because PSE has 746 MW of Colstrip transmission rights and has executed a memorandum of understanding to purchase up to a 25% (~750 MW) share of the proposed North Plains Connector, which PSE is evaluating as a transmission sensitivity rather than including in its reference case (June 23 RPAG, slides 30 and 55). Grid-enhancing technologies such as dynamic line ratings, advanced reconductoring, and power-flow control can unlock further capacity on existing paths, as FERC Order 1920 directs planners to consider (Request 7). Reference B's own 3,100 MW of wind and 1,275 MW of solar sit behind the same WA East limits, so these transmission limits constrain PSE's preferred plan at least as much as ours. PSE's own June 23 analysis quantifies that scale: it puts total transmission need at 4,497 MW in 2035, rising to 7,039 MW by 2045, and finds that delivering its preferred renewables across the Cascades requires at least two new 500 kV lines—about 3,500 MW of added capacity—on the West of Cascades North path (June 23 RPAG, slides 28–29 and 58). Increasing the limits benefits any portfolio PSE selects.

Appendix B: Limitations

To model behavior we wish to see from PSE, let us state the weak points of our plan plainly.

The clean electric planning case covers **85% of the gas and nuclear it replaces at 2035 (93% of the full no-gas need)**, not 100%. The residual can come from the contingency block, from achievable-case performance, or, if every milestone slips, from a smaller, deferred thermal decision.

V2G seems very likely to become common well before 2050, but we are currently early in its use. Correspondingly, our planning case attributes only 100 MW to V2G in 2035.

The pace of electrification is uncertain. It depends on the pace of building conversion, on participation in flexible-load programs, and on weatherization. Our electrification load and cost figures are estimates that need formal modeling.

The combined-cost comparison rests on PSE's gas-supply figure plus a speculative reconstruction of total gas costs and avoided allowances. It blends utility system cost with customer-side capital, which may be unconventional. The residual gas figure is scenario-based rather than derived, and the error bands are wide and overlap.

We frame the **HB 1589** point as a planning-consistency challenge PSE must answer, not a finding of legal violation. Initiative 2066's amendments and the pending Supreme Court decision are addressed in the text, and our requests are modeling and disclosure requests that no reading of I-2066 prohibits. The electrification peak is sensitive to how far conversions rely on cold-climate equipment rather than resistance backup.

Our memo does not prove that **zero new thermal capacity** will ever be needed. It shows that PSE has not justified the scale, timing, and exclusivity of the gas buildout it proposes, on either the electric or the gas side of its own plan.

One problem this plan does not solve is **PSE's existing gas generation**. Keeping new gas off the electric system and winding down the retail gas network still leaves PSE's current gas power plants in service (1,779 MW of effective winter peak contribution, per PSE's slide 24). Those plants require a maintained, if much smaller, gas supply to operate, and they continue to emit. The Clean Energy Transformation Act's requirement of 100 percent clean electricity by 2045 means this fleet, like PSE's proposed new gas, cannot simply keep burning fossil gas indefinitely. Fully retiring and replacing it is a further step our plan does not cost. However, not building five gigawatts of new gas keeps the problem from growing, and the diversified clean approach we describe is the same toolkit that, extended and scaled in a later planning cycle, could replace the existing fleet as well.

Appendix C: Formulas and assumptions

V2G (planning case)

Effective MW = EV stock in PSE territory × bidirectional-capable share × enrollment rate × kW per vehicle × winter coincidence factor. 2035: $220,000 \times 25\text{--}30\% \times 50\% \times 5 \text{ kW} \times 65\% \approx 90\text{--}110 \text{ MW}$ (planning 100). 2045: $460,000 \times 65\% \times 45\% \times 6 \text{ kW} \times 65\% \approx 525 \text{ MW}$ (planning 500). 2050: $590,000 \times 70\% \times 50\% \times 6 \text{ kW} \times 65\% \approx 805 \text{ MW}$ (planning 800). EV stock assumes Washington's current ~286,000 registered plug-in vehicles (~230,000 of them battery-electric, with the bidirectional-capable share applied to the full plug-in fleet) growing 4–10%/yr, PSE territory ~42% of the state fleet. Enrollment assumes a bidirectional-charger incentive funded at an avoided-peaker carrying cost derived as PSE's \$1,624/kW peaker capital × ~15% annual carrying charge ≈ \$245/kW-yr (conservative against PSE's draft leveled capacity-cost benchmarks of \$383–1,136/kW-yr, slide 88) × 5 kW ≈ \$1,200/vehicle-yr, consistent with Connecticut's pilot, which pays up to \$1,350/yr. Regulators should treat the planning case, not the achievable case, as the burden-of-proof benchmark.

Demand response, BTM VPPs, substation storage, iron-air, wind, geothermal

DR: PSE states the cumulative 10% demand-response target “could be possible” (slide 19). Applied to the 5,650 MW 2050 no-gas deficit, a conservative base since 2050 winter peak demand is larger, that implies ~565 MW. The planning case applies a 20% accreditation and enrollment discount (≈450 MW by 2035), counting only contractually enrolled, telemetry-verified winter capability incremental to the 210 MW baseline. Managed V1G charging is included here (no export), with V2G shown separately. BTM VPPs: nameplate 200/400/500 MW × 80% dispatch × 85% credit. Substation storage: 600/700/800 MW of 5 MW/20 MWh clusters × 85% × 90%, with capital priced at PSE's Li-ion entry. For \$/kW-yr comparisons we use PSE's DER RFP Bands 2–3 (\$204–252/kW-yr, since Band 1's \$114/kW-yr reflects only ~5 MW of RFP depth and utility-scale procurement at hundreds of MW should sit at or below the higher bands). Interconnection is omitted on both sides, partly offset by non-wires benefits. PSE's June 23 transmission reference case puts the local-network-upgrade cost of integrating 3,000 MW of resources at ~\$1.5B (slides 24–25); applied symmetrically, a share would attach to the utility-scale in-zone additions here (iron-air, geothermal, substation storage), while the distribution-connected majority of this portfolio would not trigger transmission network upgrades. Iron-air: 350/1,200/2,000 MW × 95% winter availability at PSE's \$2,273/kW (slide 81 “Long Duration Storage Iron Air,” available 2035, with the 100-hour duration being the vendor specification in PSE's announced pilot), one of several long-duration substitutes. Wind: +500/750/1,000 MW × 35% winter credit. Geothermal: 100/167/444 MW × 90% (EGS available 2035 at \$11,893/kW vs. SMR \$13,518/kW at 2040). PSE's portfolio row is labeled “Nuclear,” with slide 81 identifying the modeled technology as Nuclear SMR.

Full-electrification scenario

Residential gas homes converting ≈ 800,000 (of ~900,000 gas customers). Added winter peak per home at a cold-snap hour ≈ 3.0–4.5 kW electric (mid 3.5), reflecting a cold-climate heat pump holding capacity near 5°F at a minimum COP ~2.1 with limited resistance backup. Gross ≈ 2,800 MW. Weatherization/efficiency ≈ –25% (≈ –700 MW). Self-managed flexibility ≈ –45% of the post-efficiency peak (HPWH thermal storage ~960 MW + heat-pump pre-heat ~640 MW ≈ 1,600 MW of new flexibility, ~950 MW credited net of the weatherization step). Net new firm capacity ≈ 1,150 MW, met by +500 iron-air, +300 winter wind, +200 geothermal, +150 substation/BTM. Delivered energy per converted home falls ≈ 60% (gas furnace ~95% efficient vs. heat pump seasonal COP ~2.5). Our figures are speculative, to be refined by modeling a high-electrification scenario.

Combined electric + gas cost (speculated)

PSE path: electric Reference B \$55.4B (PSE direct). Gas plan ~\$70B, reconstructed from PSE's gas-supply NPV of \$29.2B (2028–2050, PSE direct) scaled by gas supply's ~42% average share of total gas cost over the period (PSE breakdown: supply 51%→34%, T&D 25%→14%, CCA 22%→51%, 2031→2050). Embedded gas CCA ≈ \$24B. These CCA figures inherit PSE's own ceiling-price assumption (slide 87 applies the CCA ceiling forecast to both fuels). At recent settlement prices (~\$65/ton, March 2026) the burden shrinks but remains the fastest-growing component of PSE's own 2031→2050 cost breakdown. Combined ≈ \$125B. WCEC path: scaled clean electric ~\$59.5B (base ~\$55B + ~\$4.5B for the +1,150 MW electrification firm capacity), electrification capital ~\$13B (~\$16k/home incremental above business-as-usual end-of-life appliance replacement × 800,000, range \$10–16B), and residual gas represented by a mid-range scenario of ~\$28B within a broader ~\$20–40B range. This is not an erasure of the existing system's embedded cost: it counts continued recovery of undepreciated gas assets plus a declining tail of supply and operations, while avoiding the large forward investment in new mains and the escalating CCA burden that PSE's path carries. Reaching the lower end assumes the Commission authorizes securitization (refinancing the stranded assets at lower bond rates rather than PSE's authorized rate of return) and accelerated depreciation. Without those mechanisms, the residual sits toward the top of the range. Combined ≈ \$100B. Error bands are wide (PSE ≈ \$115–130B, WCEC ≈ \$90–115B) and overlap. The comparison turns on how much of the gas CCA burden is avoided. Whole-home electrification typically runs \$25,000–40,000 before rebates (larger homes run higher). The incremental figure used here counts only the cost above the furnace, air-conditioner, and water heater a household would replace anyway over 25 years. Customer operating-bill savings (~60% per converted home) and rebates (Washington's state HEAR program, operating now, and the federal Home Energy Rebates of up to ~\$14,000 for income-eligible households, pending launch in Washington) are not netted into the capital figures above.

Retained Reference B non-gas builds at 2045 (~2,705 effective MW)

Utility-scale storage ~2,000 MW × 85% ≈ 1,700 MW. Wind ~2,700 MW × 35% ≈ 945 MW. Solar ~1,200 MW × 5% winter credit ≈ 60 MW. Total ≈ 2,705 MW, reflecting this memo's speculative accreditations applied to PSE's own announced builds. No resource in this subtotal appears elsewhere in the portfolio table. At 2035 the same accreditations give roughly 2,040 MW retained (storage 1,394 + wind 595 + solar 49), making the 2035 no-gas need roughly 1,924 + 2,040 = 3,960 MW. At 2050 they give roughly 2,900 MW against the 5,650 MW need on slide 24. A stricter lens divides our additions alone by the gas and nuclear they replace: 1,924 MW at 2035 and 5,342 MW at 2050 (peakers plus the SMR). The planning case reaches 85% and 89% by that measure, the achievable case 104% and 109%. The main text reports the whole-portfolio measure because the whole system is what serves customers. Both lenses are shown here for completeness.

Electric portfolio cost (~\$55B) and externalities

From Reference B's \$55.4B, remove peaker capital (\$8.2B), SMR (\$4.1B), and avoided peaker operating costs (~\$3.7B NPV: fixed O&M \$25/kW-yr, fuel at 8% CF, 9.5 MMBtu/MWh, and \$5/MMBtu, with CCA at the \$80/ton ceiling). Add iron-air (\$4.5B), substation storage (\$1.5B), wind (\$2.0B), geothermal (\$5.3B), chargers (\$0.6B), and program payments (~\$2.0B). Net ≈ \$0 vs. Reference B (cost parity). All unit capital costs are PSE's own, and they already reflect the 2025 federal repeal of clean-energy tax credits (slide 75), so this portfolio is priced without expired subsidies. Peaker capital uses PSE's \$1,624/kW "Natural Gas Peaker" entry. PSE prices a "Renewable Peaker" at \$1,988/kW, so if Reference B's flex-fuel builds carry the higher cost, peaker capital is ~\$10.0B and the first tranche ~\$3.8B. Our use of the lower figure is conservative. It is doubly conservative against the market: BNEF reports that U.S. gas plant

construction costs rose 66% from 2023 to 2025, with time to bring plants online up 23%. Externalities: $5,042 \text{ MW} \times 8\% \text{ CF} \times 8,760 \text{ h} \times 0.50 \text{ tCO}_2/\text{MWh}$ ($9.5 \text{ MMBtu}/\text{MWh} \times 53 \text{ kg CO}_2/\text{MMBtu}$) $\approx 1.8 \text{ Mt/yr}$. At the social cost of carbon (EPA's 2023 peer-reviewed central estimate of \$190/ton at a 2% discount rate, no longer used federally but consistent with the social cost of greenhouse gases Washington law applies in utility analysis, RCW 80.28.405) less the \$80 internalized via CCA, roughly \$195M/yr is externalized beyond the allowance price. The geothermal-for-SMR swap also avoids HALEU supply and spent-fuel and decommissioning liabilities not in PSE's SMR capital cost.

Sources

1. PSE RPAG presentation, May 28, 2026. Slides relied on: 18–19, 24, 28–31, 33, 44, 50–51, 59, 75, 81, 84, 87–88, 104, 119. Covers Reference Cases A/B, gas rate impacts of +64/113/392% for 2030/2035/2050 (which PSE labels illustrative and based on CCA ceiling prices), CCA growing from 22% to 51% of gas cost, “miles of main remains constant,” and targeted electrification limited to “3–5 small projects annually” and the Duvall pilot. RPAG materials are drafts subject to change, so we retain an archived copy. https://irp.cdn-website.com/dc0dca78/files/uploaded/2026_0528_RPAG_Meeting_Final1.pdf
2. PSE RPAG presentation, June 23, 2026. Slides relied on: 10, 14–18, 22–25, 28–32, 55–58. Covers the electric distribution system reference case (~\$4.1B of distribution investment 2026–2050, substation overloads, and 5 MW BESS co-location as a non-wires alternative), the local and regional transmission reference cases (PSE’s 746 MW Colstrip rights, an MOU for up to 25% / ~750 MW of the North Plains Connector, ~\$1.5B of local network-upgrade cost to integrate 3,000 MW of resources, and total transmission need of 4,497 MW in 2035 rising to 7,039 MW by 2045), and PSE’s recap framing the infeasible initial electric reference case as evidence of system risk. RPAG materials are drafts subject to change, so we retain an archived copy. https://irp.cdn-website.com/dc0dca78/files/uploaded/2026_0623_RPAG_Meeting_Final_V1.pdf
3. Washington HB 1589 (2024): combination utility decarbonization and Integrated System Plan requirement (House Bill Report, as passed Legislature). As amended by Initiative 2066 (2025 c 1). See current RCW 80.86.020 (source 19) and the pending Washington Supreme Court review, argued January 22, 2026. <https://lawfilesexternal.wa.gov/biennium/2023-24/Pdf/Bill%20Reports/House/1589-S.E%20HBR%20PL%2024.pdf>
4. PSE, “Facts about HB 1589” (≈900,000 gas customers, combination utility). <https://www.pse.com/en/press-release/details/Facts-about-HB-1589>
5. Rewiring America: upfront cost of home electrification (typical whole-home packages roughly \$25–40k before rebates). For federal Home Energy Rebate status in Washington, see WA Dept. of Commerce. <https://www.rewiringamerica.org/research/home-electrification-cost-estimates>
6. Evolved Energy Research / PNNL: cold-climate heat pump winter peak and electrification load shape. <https://www.evolved.energy/posts/planning-the-grid-with-next-gen-cold-climate-heat-pumps>
7. FERC, January 2026 Commission meeting summaries (Goldendale license, P-14861). If the page does not render, see the order in Docket P-14861 or contemporaneous news coverage. <https://www.ferc.gov/news-events/news/summaries-january-2026-commission-meeting>
8. Columbia Riverkeeper / Yakama Nation challenges to Goldendale license. <https://www.columbiariverkeeper.org/2026/license-to-destroy-sacred-tribal-resources-challenged/>
9. PSE announcement: PSE partners with Form Energy on a 10 MW / 100-hour iron-air pilot (January 2024, MOU stage, targeted for late 2026). <https://www.pse.com/en/press-release/details/Puget-Sound-Energy-partners-with-Form-Energy-to-evaluate-multi-day-energy-storage-solutions>
10. Utility Dive: Xcel Energy / Google 300 MW / 30 GWh iron-air project (announced February 2026, first deliveries expected 2028, largest announced to date by energy capacity). <https://www.utilitydive.com/news/worlds-largest-grid-battery-part-of-google-xcel-energy-agreement/813793/>
11. Brattle Group, Real Reliability: The Value of Virtual Power (2023). Finds VPPs deliver resource adequacy at roughly 40–60% of the net cost of a gas peaker. <https://www.brattle.com/insights-events/publications/real-reliability-the-value-of-virtual-power/>
12. Washington EV registration data, data.wa.gov (285,822 plug-in vehicles including ~56k PHEVs, June 2026). <https://data.wa.gov/Transportation/Electric-Vehicle-Population-Data/f6w7-q2d2>
13. WA Dept. of Ecology, CCA auction results (March 2026 auction settled at \$65.26/ton, June 3, 2026 auction results pending at this writing, \$80 ceiling for 2026–27 per HB 1975, and CA–QC linkage still in draft with a linked market expected no earlier than 2027). <https://ecology.wa.gov/air-climate/climate-commitment-act/cap-and-invest/auctions-and-market>
14. Connecticut residential V2G pilot compensation (up to \$1,350/yr with full event participation, 63-vehicle pilot). <https://evclubct.com/states-first-vehicle-to-grid-pilot-enables-ev-owners-to-get-paid-to-support-the-grid/>
15. PSE Flex Batteries program (existing VPP precedent). <https://www.pse.com/en/rebates/PSE-flex/flex-batteries>
16. Energy Institute at Haas WP 343: advanced conductors in existing right-of-way. <https://haas.berkeley.edu/wp-content/uploads/WP343.pdf>

17. Priyadarshan, Crozier, Baker & Kircher, “Distribution grids may be a barrier to residential electrification,” Cell Reports Sustainability (2025): per-household distribution reinforcement of \$2,800–6,400, over two-thirds avoidable with demand-side management. [https://www.cell.com/cell-reports-sustainability/fulltext/S2949-7906\(25\)00214-9](https://www.cell.com/cell-reports-sustainability/fulltext/S2949-7906(25)00214-9)
18. Western Power Pool, Assessment of January 2024 Cold Weather Event (January 12–16, 2024, record Northwest winter peak near 35,500 MW, sustained multi-day imports averaging ~4,900 MW). https://www.westernpowerpool.org/private-media/documents/WPP_Assessment_of_January_2024_Cold_Weather_Event_Final.pdf
19. Washington UTC, integrated system plan rules adopted 2025 (first PSE ISP due 2027). <https://www.utc.wa.gov/news/2025/state-regulators-adopt-rules-integrated-system-plans>
20. RCW 80.86.020 as amended by Initiative 2066 (2025 c 1). I-2066 was ruled unconstitutional by King County Superior Court (March 21, 2025) and remains in effect pending Washington Supreme Court review (argued January 22, 2026). <https://app.leg.wa.gov/RCW/default.aspx?cite=80.86.020>
21. DOE Residential Cold Climate Heat Pump Challenge: prototypes deliver 100% heating capacity at 5°F at COP ≥ 2.1. <https://www.energy.gov/eere/buildings/residential-cold-climate-heat-pump-challenge>
22. Goldendale Energy Storage Project license application, economics study: construction cost approximately \$2.5–2.9 billion (2019 dollars) for 1,200 MW. https://www.ryedevlopment.com/assets/documents/goldendale/FLA/Goldendale-FLA_Appendix-I_2019-Economics-Study-Report.pdf
23. E3, Resource Adequacy and the Energy Transition in the Pacific Northwest (Phase 1 September 2025, final report April 2026): projects a roughly 9 GW regional gap by 2030. <https://www.ethree.com/ra-pnw/>
24. Hart, Sylvan Energy Analytics, independent review of the E3 Northwest resource adequacy study, sponsored by GridLab (January 2026): large-load flexibility could mitigate most or all near-term winter needs under most load scenarios. https://www.sylvan.energy/s/NW_RA_Study_Review_Final.pdf
25. Northwest Power and Conservation Council, Pacific Northwest Power Supply Adequacy Assessment for 2029 (document 2024-4): the region is adequate through 2029 if high-end efficiency, roughly 6,600 MW of renewables, and demand response are delivered. <https://www.nwcouncil.org/reports/2024-4/>
26. KUOW (January 14, 2024): complete outage at PSE’s Jackson Prairie gas storage facility during the January 2024 cold snap, with PSE urging customers to conserve gas and electricity. <https://www.kuow.org/stories/why-pse-urged-people-to-conserve-energy-amid-severe-cold>
27. Bloomberg / BloombergNEF (April 2026): U.S. gas plant construction costs rose 66% from 2023 to 2025, with time-to-online up 23%. <https://www.bloomberg.com/news/articles/2026-04-23/cost-to-build-natural-gas-fired-power-plant-surges-66-bnef-says>
28. Washington EFSEC, facilities list (<https://efsec.wa.gov/facilities>): the only operating EFSEC-certified gas plants are Chehalis (2003) and Grays Harbor (2008), and every facility now under review is solar, battery, wind, or transmission. The last EFSEC-permitted gas projects—[Sumas Energy 2](#) (certification terminated 2006), [BP Cherry Point Cogeneration](#), and [Wallula Power Project](#)—were cancelled by their developers and never built.
29. RCW 19.405.040(8): utilities must ensure the equitable distribution of energy and nonenergy benefits and reduction of burdens to vulnerable populations and highly impacted communities. <https://app.leg.wa.gov/RCW/default.aspx?cite=19.405.040>
30. Krieger, Casey and Shonkoff, Energy Policy (2016): for some California peakers, over one third of generation occurs on days that already violate federal air standards. <https://www.sciencedirect.com/science/article/abs/pii/S0301421516302798>
31. PEAK Coalition, Dirty Energy, Big Money (2020): New York City peakers are overwhelmingly sited in environmental-justice communities (illustrative of siting-equity dynamics, not Washington-specific). <https://www.cleaneogroup.org/wp-content/uploads/Dirty-Energy-Big-Money.pdf>
32. Washington State Standard (December 9, 2025): TransAlta and PSE agreement to convert the existing Centralia coal plant to gas (700 MW, approvals and permitting pending). <https://washingtonstatestandard.com/2025/12/09/washingtons-last-coal-power-plant-will-transition-to-natural-gas/>
33. RCW 19.280.020: definition of “lowest reasonable cost,” which expressly includes “the risks imposed on the utility and its ratepayers.” <https://app.leg.wa.gov/RCW/default.aspx?cite=19.280.020>

34. U.S. DOE, HALEU Frequently Asked Questions and HALEU Availability Program (domestic enrichment ~900 kg/yr, projected need above 40 metric tons by 2030). <https://www.energy.gov/ne/haleu-frequently-asked-questions>
35. Krall, Macfarlane and Ewing, "Nuclear waste from small modular reactors," PNAS (2022): 2 to 30 times more waste per unit energy. An industry comment and author reply are published alongside. <https://www.pnas.org/doi/10.1073/pnas.2111833119>
36. White House fact sheet (May 23, 2025): executive order directing NRC reorganization, wholesale rule revision, and 18-month licensing deadlines. <https://www.whitehouse.gov/fact-sheets/2025/05/fact-sheet-president-donald-j-trump-directs-reform-of-the-nuclear-regulatory-commission/>
37. CNBC (July 2025): former NRC officials warn the overhaul weakens the regulator. Commissioner removed June 2025, more than 400 staff departures in 2025. <https://www.cnbc.com/2025/07/17/trumps-nuclear-power-push-weakens-regulator-and-poses-safety-risks-former-officials-warn.html>
38. NuScale and UAMPS, joint announcement terminating the Carbon Free Power Project (November 2023) after the target price rose from \$58 to \$89/MWh with ~\$1.4B in federal support. <https://www.sec.gov/Archives/edgar/data/0001822966/000182296623000256/uampsnuscalejointpressre.htm>
39. World Nuclear News (2025): OPG Darlington BWRX-300 budget, C\$20.9 billion for four units totaling 1,200 MW. <https://www.world-nuclear-news.org/articles/what-is-the-budget-for-canadas-first-smr-project>
40. Lazard, LCOE+ (June 2025): unsubsidized nuclear \$141 to 251/MWh, geothermal \$66 to 109/MWh. <https://www.lazard.com/media/eijnqja3/lazards-lcoeplus-june-2025.pdf>
41. Fervo Energy (March 2026): \$421 million non-recourse project financing for Cape Station, 500 MW under utility PPAs, drilling costs down by half. <https://fervoenergy.com/fervo-energy-secures-421-million-in-non-recourse-project-financing-for-cape-station/>
42. NREL, Enhanced Geothermal Shot analysis (NREL/TP-5700-84822): ~106 GW of EGS potential in the Cascades region, DOE cost target \$45/MWh by 2035. <https://www.osti.gov/biblio/1922621>
43. PSE 2023 Electric Progress Report, Chapter 7, Table 7.19: nuclear ELCC of 99%. No geothermal ELCC has been published. https://www.pse.com/-/media/PDFs/IRP/2023/electric/chapters/07_EPR23_Ch7_Final.pdf
44. POWER Magazine: PSE commits \$10 million to Energy Northwest SMR feasibility work, X-energy Xe-100 the leading contender. <https://www.powermag.com/puget-sound-energy-commits-10m-to-nuclear-power-feasibility-work-x-energy-smr-technology-a-leading-contender-for-project/>
45. Davis and Hausman, "Who Will Pay for Legacy Utility Costs?" JAERE (2022): a 10% customer loss reduces revenues only ~5%, with remaining customers absorbing the difference. <https://www.nber.org/papers/w28955>
46. E3 for the California Energy Commission, "The Challenge of Retail Gas in California's Low-Carbon Future" (2020): declining demand raises rates and "a feedback loop takes effect." <https://www.energy.ca.gov/sites/default/files/2021-06/CEC-500-2019-055-F.pdf>
47. Gridworks, "California's Gas System in Transition" (2019): residential gas rates modeled rising toward \$19/therm by 2050 under high electrification. <https://gridworks.org/initiatives/cagas-system-transition/>
48. NW Natural Holdings Form 10-K (2024): names propane among its primary competitors and identifies customer-retention risk when gas prices are high. <https://www.sec.gov/Archives/edgar/data/0001733998/000173399825000020/nwn-20241231.htm>
49. EPA GHG Emission Factors Hub (40 CFR Part 98, Table C-1): propane 62.87 vs. pipeline natural gas 53.06 kg CO2/MMBtu. <https://www.epa.gov/climateleadership/ghg-emission-factors-hub>
50. Washington E2SHB 2215 (2026): lowers the CCA fuel-supplier threshold to 500 tons for suppliers formed after January 1, 2023, responding to threshold gaming. <https://app.leg.wa.gov/billsummary/?billNumber=2215&year=2026&initiative=False>
51. WA UTC (2025): PSE rate orders (gas +10.6% January 2025, ~+7% January 2026) and the Docket U-230968 CCA risk-sharing order. <https://www.utc.wa.gov/news/2025/state-regulators-approve-multiple-orders-impacting-puget-sound-energy-rates>
52. Clean Energy Transition Institute, "How the Northwest Heats Homes" (2024): about 4% of Northwest homes heat with propane. <https://www.cleanenergytransition.org/post/how-the-northwest-heats-homes>

53. Seattle Times Climate Lab (2024): PSE residential gas deliveries fell ~7% from 2022 to 2023. <https://www.seattletimes.com/seattle-news/climate-lab/puget-sound-energy-proposes-rate-hikes-to-pay-for-hydro-and-wind-power/>
54. S&P Global Ratings (2021): long-term credit quality for some gas distribution utilities will become “increasingly challenging.” <https://www.spglobal.com/ratings/en/regulatory/article/210428-how-esg-factors-are-shaping-north-american-regulated-investor-owned-utilities-credit-quality-s11866635>
55. CBS Los Angeles (2024): after SoCalGas shut off service in Rancho Palos Verdes, residents switched to propane. <https://www.cbsnews.com/losangeles/news/socalgas-shuts-off-natural-gas-service-to-rancho-palos-verdes-due-to-worsening-land-movement/>